

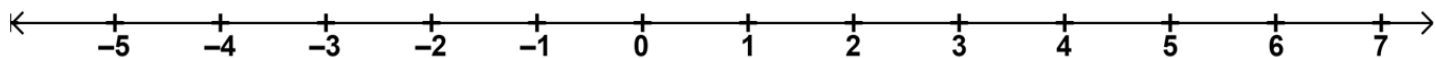
Solve each inequality. Show your work.

1. $-11.3 > -3x - 0.5$	2. $\frac{9}{5}x + 23 \leq -58$
Solution	Solution
3. $\frac{2}{3}(21 - 15x) > 76$	4. $6.2(x - 4.7) \leq 31.62$
Solution	Solution

Solve each inequality. Show your work. Graph the solution.

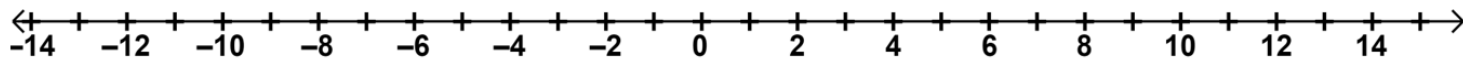
5. $9x + 7.2 > 38.7$

Solution



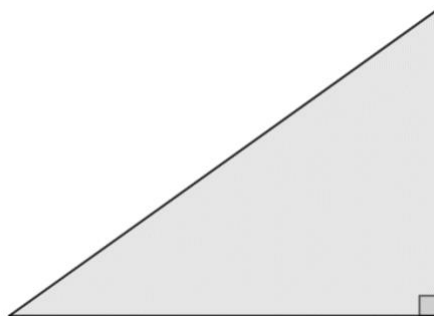
6. $-8(5x + 9) \geq 208$

Solution



7. A right triangle is shown, where the horizontal leg has a measure of 42.7 centimeters (cm) and the hypotenuse has a measure of 52.5 cm. Determine the length, in centimeters (cm), of the vertical leg. Show your work for partial credit.

Vertical Leg		cm
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8. One leg of a right triangle has a measure of 19 units. The hypotenuse has a measure of $\sqrt{7085}$ units. Determine the length, in units, of the other leg. Show your work for partial credit.

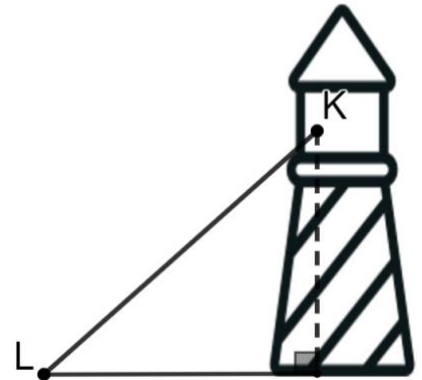
Leg		units
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9. The legs of a right triangle are congruent and have the measure of 23.6 yards. Determine the length, in yards, of the hypotenuse. Show your work for partial credit.

Hypotenuse		yards
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10. The Ponce Inlet Lighthouse in Ponce Inlet has a gallery deck where one can view the surrounding area. Kim is standing on the deck, point K , which is 140.75 feet (ft) above ground level. Louise, point L , is standing 200 ft from the base of the lighthouse directly below Kim's position, as shown.

Determine the distance, in feet, between point K and point L .



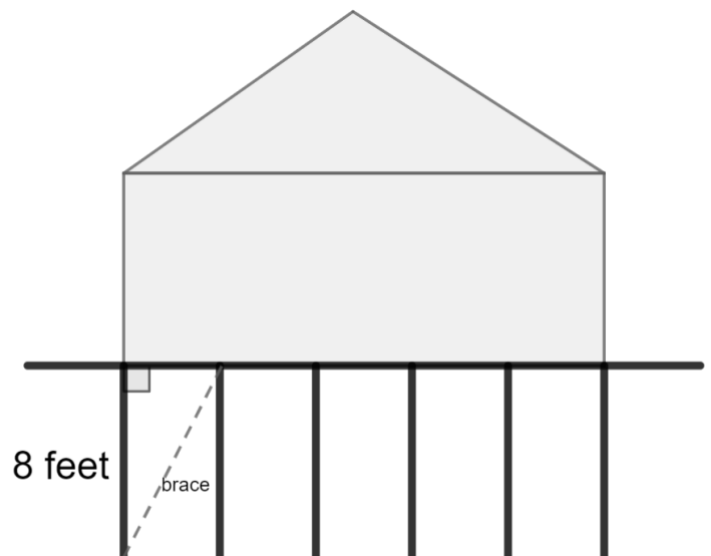
Distance between point K and point L		ft
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11. Select all of the lengths that would create a right triangle.

- ☐ 58, 156, 192
- ☐ 60, 91, 109
- ☐ 94, 168, 191
- ☐ 115, 252, 277
- ☐ 161, 240, 290

12. A house on stilts is shown where each stilt is 8 feet (ft) long. Matías built 5 braces and places each brace as shown. Each brace is 9 ft long.

Determine the length of the house, in feet.



Length of House		ft
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13. The lengths of two sides of a triangle are shown.

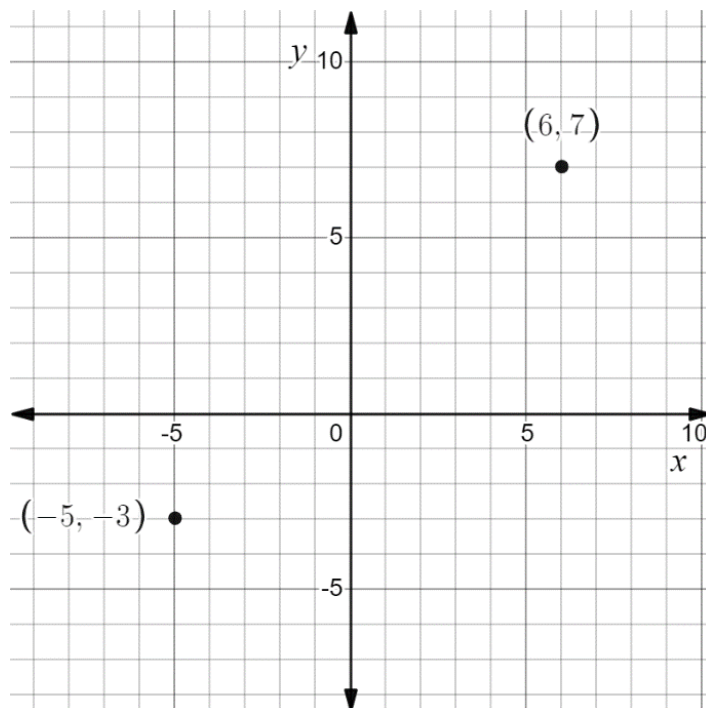
97 and 56

Complete the inequality to show the possible side measures, x , that can be used to create a triangle using the two given lengths.

Possible Side Lengths	$\underline{\hspace{2cm}} < x < \underline{\hspace{2cm}}$
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14. The points $(6,7)$ and $(-5,-3)$ are shown on the coordinate grid.

Determine the distance between the points.



Distance	
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